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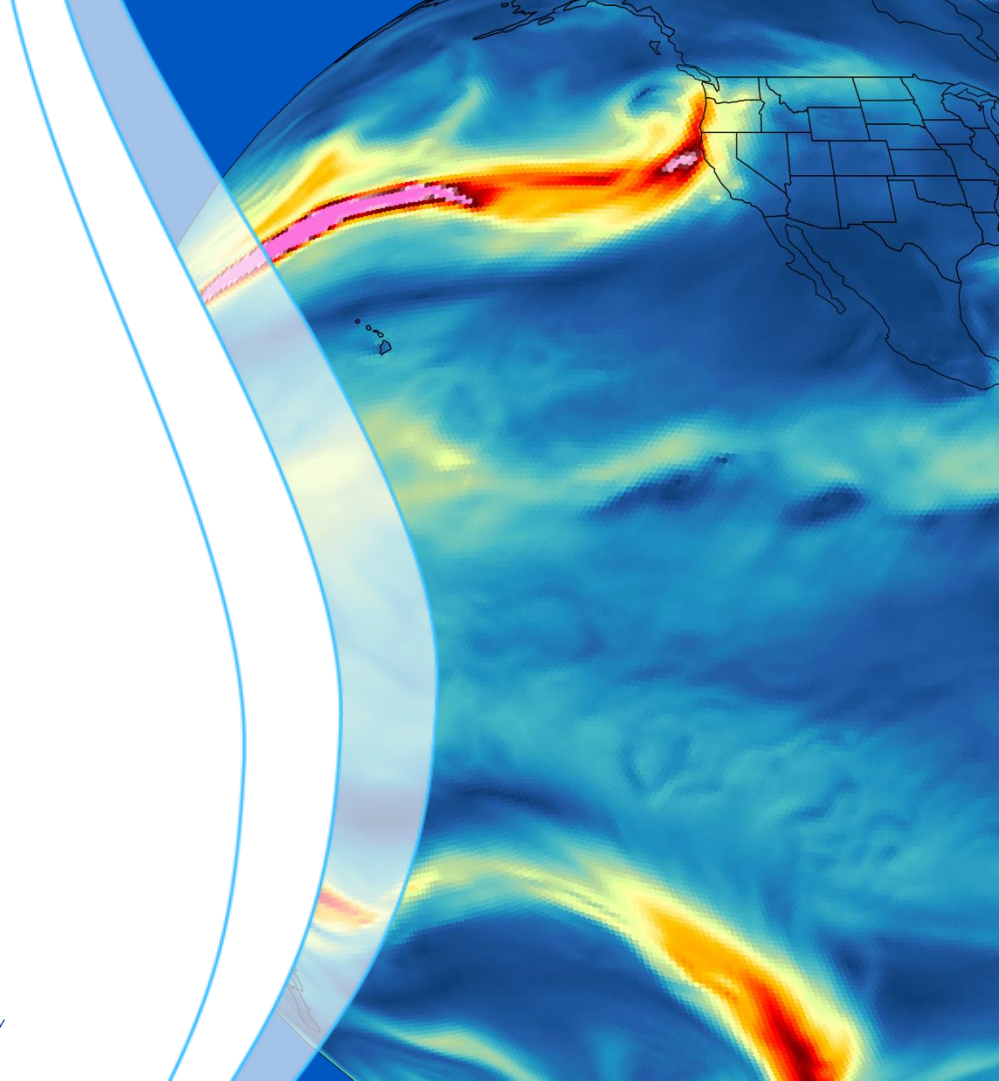
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Best Practices in Using MPAS for Weather and Forecasting Research

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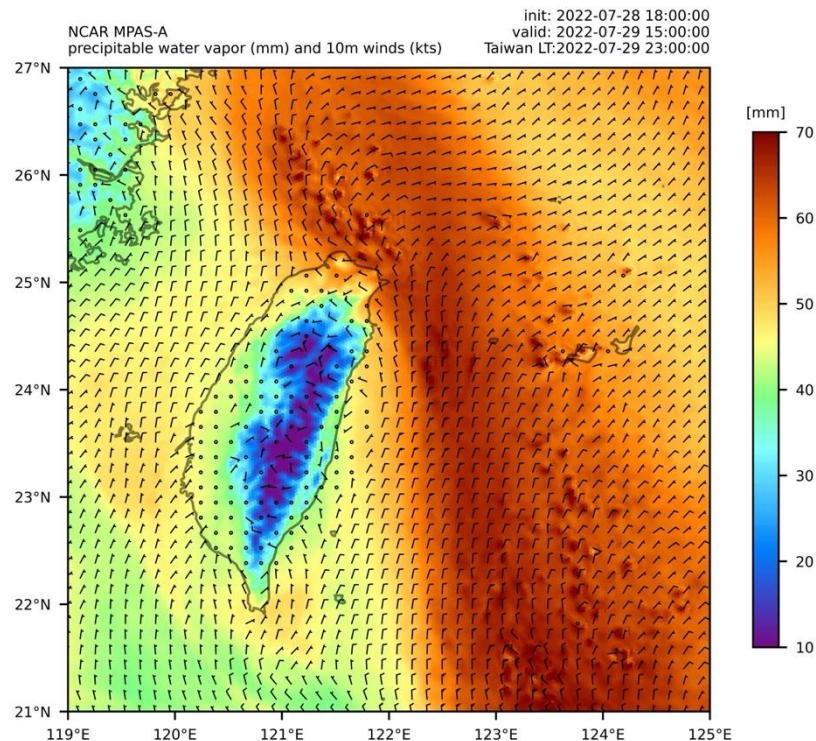


MPAS is Great!



Motivation

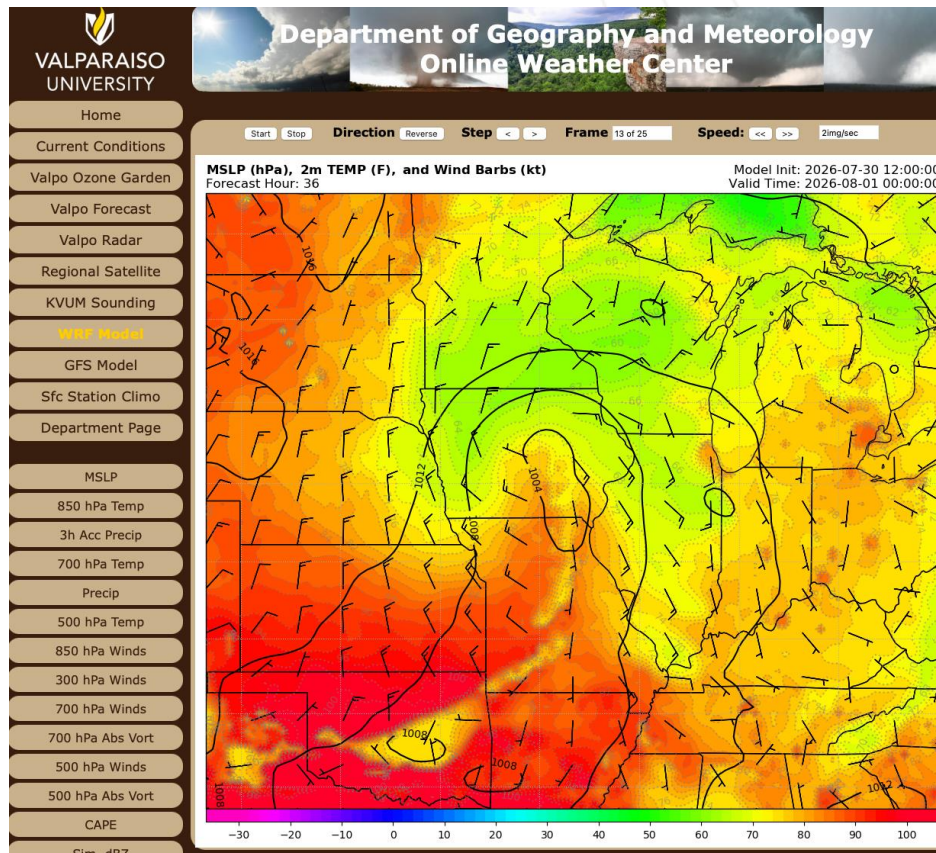
- MPAS is not so scary anymore. I promise!
- MPAS can seamlessly transition between global and regional configurations and supports both uniform and variable meshes
- I will show a comparison of parallel runs of WRF and MPAS and walk through the setup, postprocessing and analysis
- I will also cover some of the tools that are available to make a transition from WRF to MPAS as seamless as possible
- Also! If you do want to regrid your beautiful polygons to a lat/lon grid, that's easy too!



Valparaíso WRF Runs

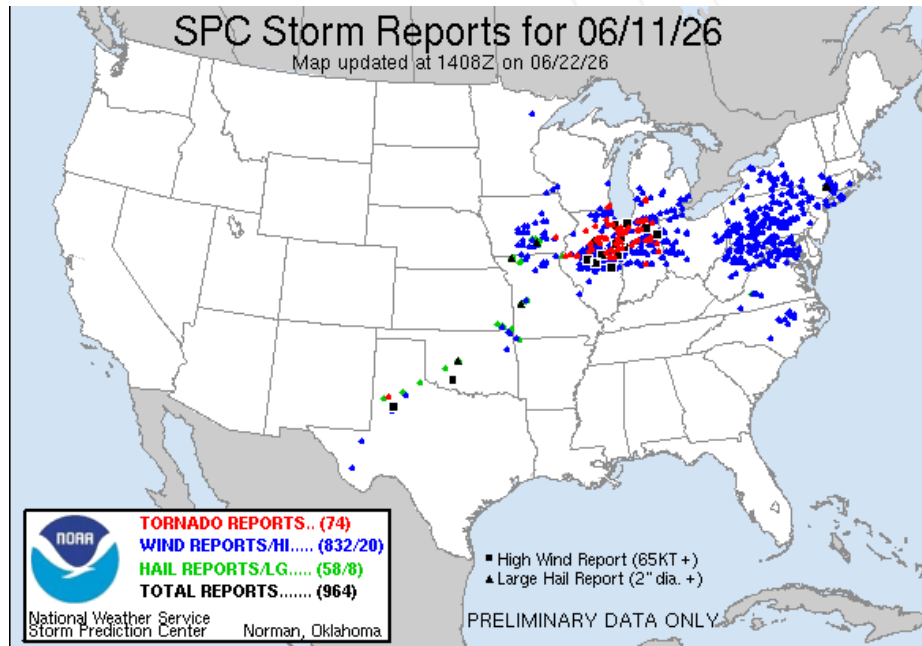
WRF runs

- Valparaíso University runs a 20km 3-day forecast run each day over a Midwestern domain
- Simple run! Let's try to do it with a similar domain using MPAS and see how they compare
- The university teaches students to run WRF.
What about MPAS?
 - Easier to set up
 - Easier to analyze
 - Just as easy to visualize



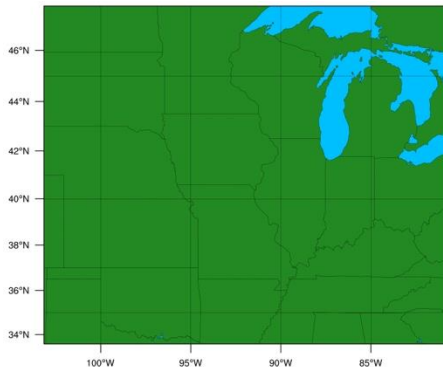
WRF and MPAS runs

- Used WRF and WPS namelists from Valparaiso for WRF run
- Based on information in the WPS namelist I set up a similar sized mesh for MPAS
- Both runs are initialized on Jun 10, 2026 12Z with GFS and run for three days
 - Severe weather occurred in the center of the domain on both June 10 and June 11
 - Do the models give similar results?
- Also, for fun (!), I decided to run both over the same domain with a 3km resolution
- For MPAS, used default physics suites based on resolution.

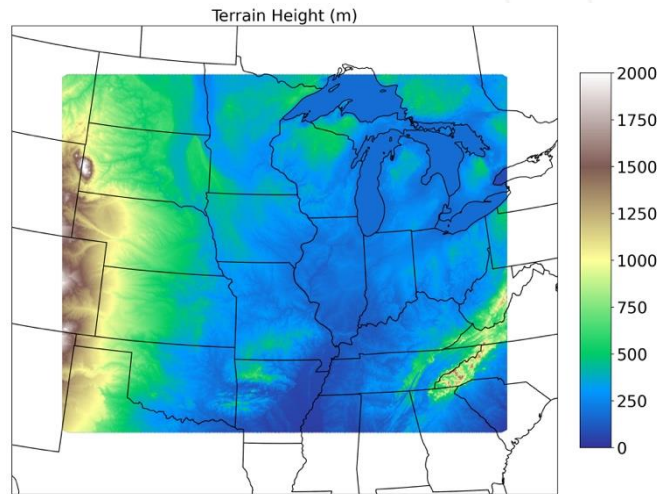


hex_projection tool in MPAS-Tools

- Easy to use tool to make a regional mesh on a lambert conformal domain
- Simple namelist that I set up based on the namelist.wps for the wrf run
- For the 3km domain, I just needed to change one line in the namelist
- Creates grid.nc and graph.info files
- MPAS-Limited-Area: you can cut out or subset part of another mesh

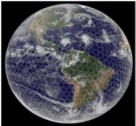


```
&mesh
  cell_spacing_km = 20.
  mesh_length_x_km = 2000.
  mesh_length_y_km = 1600.
  earth_radius_km = 6378.14
/
&projection
  projection_type = "lambert_conformal"
/
&lambert_conformal
  reference_longitude_degrees = -92.0
  standard_longitude_degrees = -92.0
  reference_latitude_degrees = 41.0
  standard_parallel_1_degrees = 30.0
  standard_parallel_2_degrees = 60.0
```



- I find setting up MPAS runs to be much simpler than WRF
- Getting input data ready:
 - Run ungrib to create WRF Intermediate files. Both models use these!
- MPAS has many more options for outputting variables you actually care about , such as **cape**, **reflectivity**, **mslp**, **temperature**, etc.
 - With WRF you have to calculate them after the fact with wrf-python or ncl
- You can easily create your own diagnostics
- You can choose which variables or diagnostics to output. You don't have these giant history files if you don't want them
- Namelist options are much easier to understand and the default physics suites work great with most runs based on your domain
- In my experience, I have way less instances with model crashes. In my experience, it somehow just usually runs and looks good!

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MPAS Atmosphere

6. Physics Suites

Documentation

- Overview of MPAS-A
- Users' Guide
- Foreword
- 1. MPAS Overview
- 2. Quick Start Guide
- 3. Building MPAS
- 4. Preparing Meshes
- Configuring I/O
- 6. Physics Suites
- Running MPAS-A

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6. Physics Suites

Beginning with version 4.0, MPAS-Atmosphere introduces a new way of selecting the physics schemes to be used in a simulation. Rather than selecting individual parameterization schemes for different processes (e.g., convection, microphysics, etc.), the preferred method is to select a **suite** of parameterization schemes that have been tested together. This is done via the namelist option `config_physics_suite` in the `&physics` namelist record. Each available suites is described in the sections that follow.

It is possible to override the choice of any individual parameterization scheme through the namelist options described in [Appendix B](#). This is useful, e.g., to disable all parameterizations except for microphysics when running some idealized simulations. See [6.4 Selecting Individual Physics Parameterizations](#) for details.

6.1 mesoscale_reference Suite

The default physics suite in MPAS-Atmosphere is the 'mesoscale_reference' suite, which contains the schemes listed in the below table. This suite has been tested for mesoscale resolutions (> 10 km cell spacing), and is not appropriate for convective-scale simulations because the Tiedtke scheme will remove convective instability before resolved-scale motions (convective cells) can respond to it.



Welcome to the MPAS tutorial practice guide

This web page is intended to serve as a guide through the practice exercises of this tutorial. Exercises are split into seven main sections, each of which focuses on a particular aspect of using the MPAS-Atmosphere model.

While going through this practice guide, you may find it helpful to have a copy of the MPAS-Atmosphere Users' Guide available in another window. Click [here](#) to open the Users' Guide in a new window.

In case you would like to refer to any of the lecture slides from previous days, you can open the [Tutorial Agenda](#) in another window.

You can proceed through the sections of this practical guide at your own pace. It is highly recommended to go through the exercises in order, since later exercises may require the output of earlier ones. Clicking the grey headers will expand each section or subsection.

0. Prerequisites and environment setup

1. Compiling MPAS, and creating static files and idealized ICs

In this section, our goal is to obtain a copy of the MPAS source code directly from the MPAS GitHub repository. We'll then compile both the `init_atmosphere_model` and `atmosphere_model` programs before using the former to begin the processing of time-invariant, terrestrial fields for use in a real-data simulation. While this processing is taking place, in another terminal window (*don't forget to set up your environment!*) we can practice preparing idealized initial conditions.

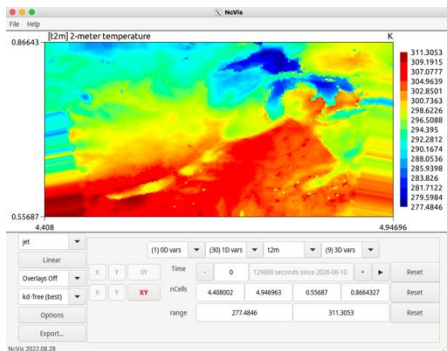
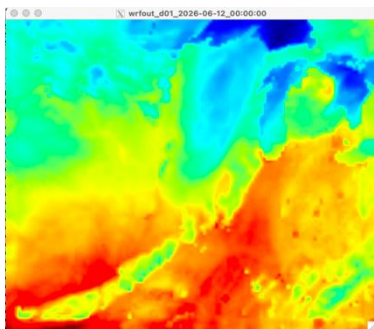
1.1 Obtaining the MPAS-Model source code



Tools for MPAS

How do I deal with the weird “grid cells”?

- How do I look at it? It doesn't work with ncview!
- **ncvis** is here to save the day! Works with unstructured meshes and easy to use
 - Just type: ncvis output.nc grid.nc
- For analysis and plotting in python **uxarray** is a great tool!
- I also heavily use **xarray** and **matplotlib** for most of my work with MPAS data



```
[1]: import uxarray as ux
```

```
[2]: mpas_20_files = "/glade/derecho/scratch/jaye/valpo/mpas_runs/20260610/rundir_20km/diag*"
grid_20 = "/glade/derecho/scratch/jaye/valpo/mpas_runs/20260610/rundir_20km/x1.11981.grid.valpo.nc"
uxds_20 = ux.open_mfdataset(grid_20, mpas_20_files, combine="by_coords")
```

```
[3]: uxds_20
```

uxarray.UxDataset

► Dimensions: (Time: 25, n_face: 11981, nVertLevels: 55, n_node: 24390, nSoilLevels: 4, nVertLevelsP1: 56)

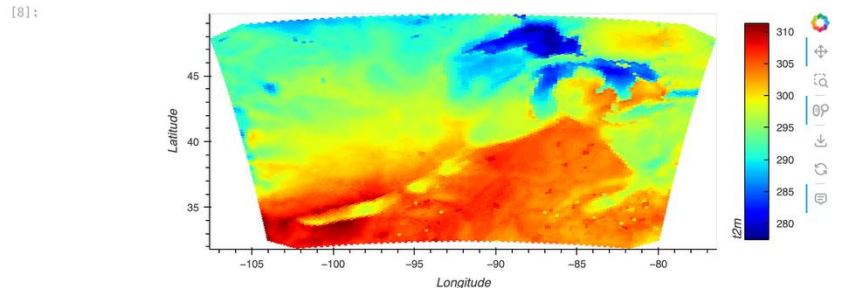
▼ Coordinates:

Time	(Time)	datetime64[ns]	2026-06-10T12:00:00 ... 2...
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▼ Data variables:

initial_time	(Time)	S64	b'2026-06-10_12:00:00 ...
xtime	(Time)	S64	dask.array<chunks=(1, ...
olrtoa	(Time, n_face)	float32	dask.array<chunks=(1, ...
rainc	(Time, n_face)	float32	dask.array<chunks=(1, ...
rainnc	(Time, n_face)	float32	dask.array<chunks=(1, ...
refl10cm	(Time, n_face, nVertLevels)	float32	dask.array<chunks=(1, ...

```
[8]: uxds_20["t2m"].sel(Time="2026-06-12 00:00:00").plot(cmap="jet")
```

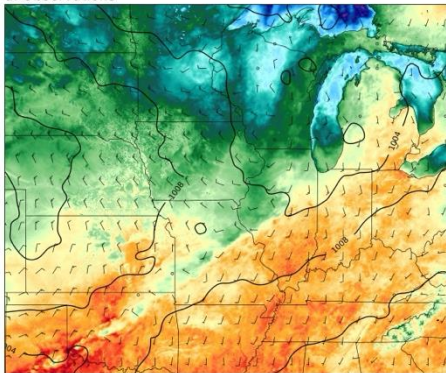


WRF vs MPAS June 11 7pm Central Time 2mTemp

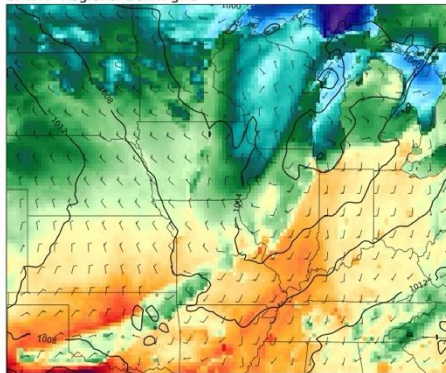


MSLP (hPa) 2m Temp (°C) and Wind (kt)

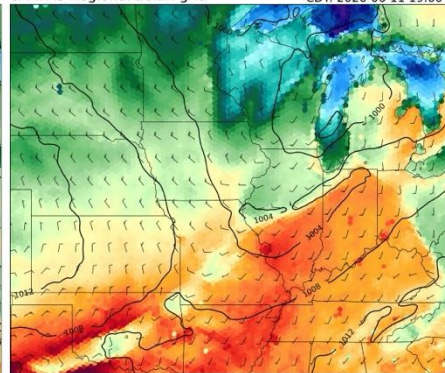
a. Observations



b. WRF Regional 20km grid



c. MPAS Regional 20km grid

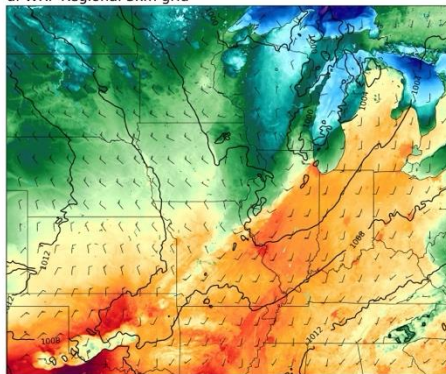


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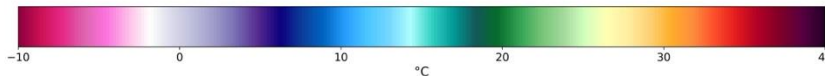
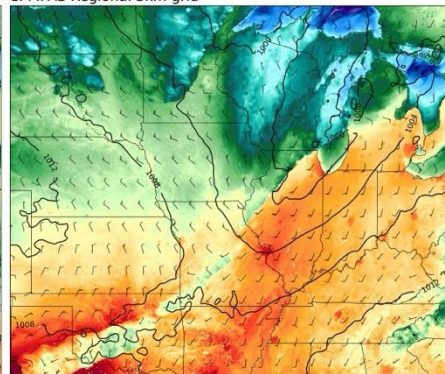
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CDT: 2026-06-11 19:00

d. WRF Regional 3km grid



e. MPAS Regional 3km grid

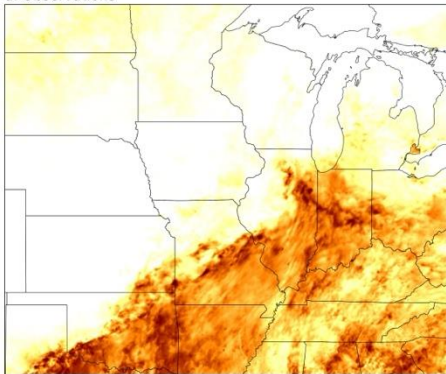


WRF vs MPAS June 11 7pm Central Time CAPE

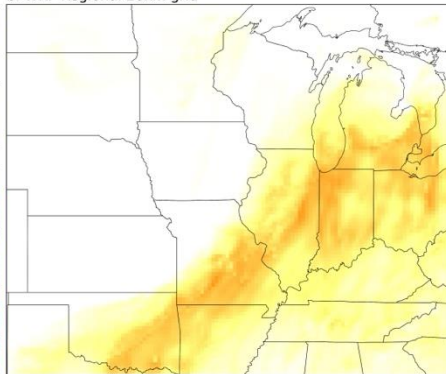


CAPE (J/kg)

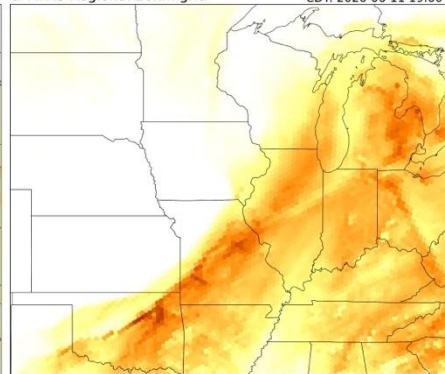
a. Observations



b. WRF Regional 20km grid



c. MPAS Regional 20km grid

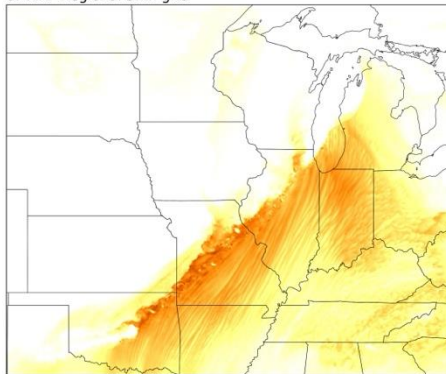


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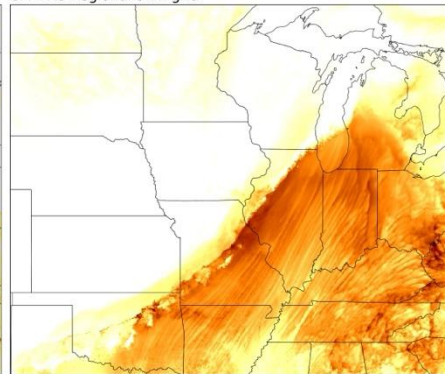
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d. WRF Regional 3km grid



e. MPAS Regional 3km grid



WRF vs MPAS June 11 7pm Central Time Max dBz

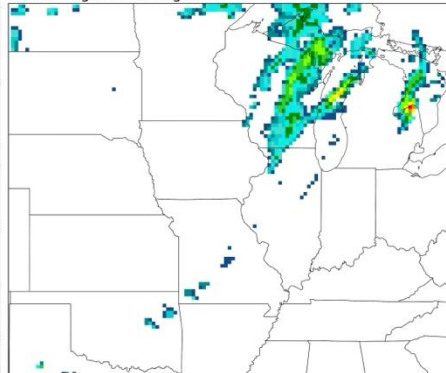


Radar Reflectivity (dBz)

a. Observations



b. WRF Regional 20km grid

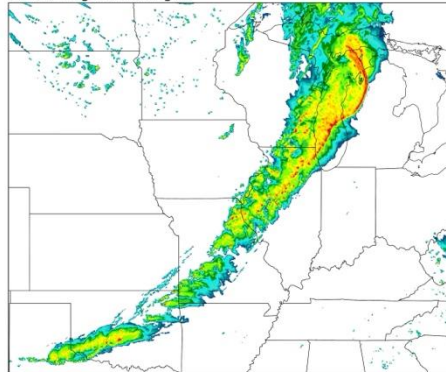


c. MPAS Regional 20km grid

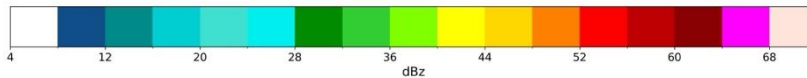
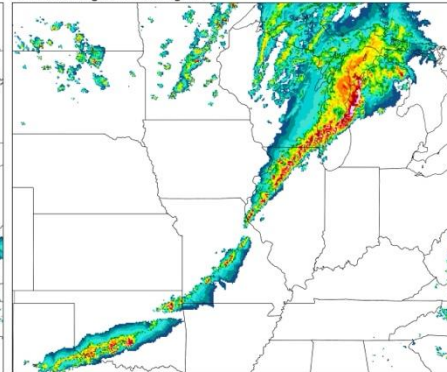


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Valid: 2026-06-12 00:00
CDT: 2026-06-11 19:00

d. WRF Regional 3km grid



e. MPAS Regional 3km grid

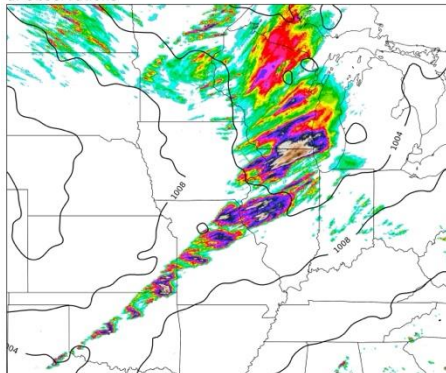


WRF vs MPAS June 11 7pm Central Time 3hr acc Prec

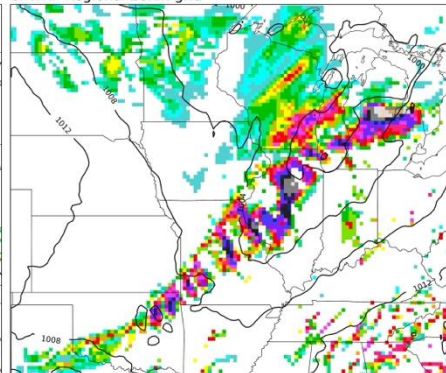


MSLP (hPa) and 3hr Accum. Precip. (mm)

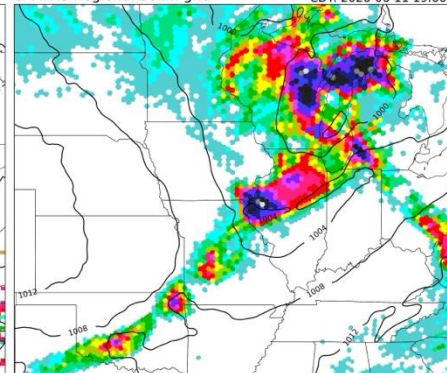
a. Observations



b. WRF Regional 20km grid

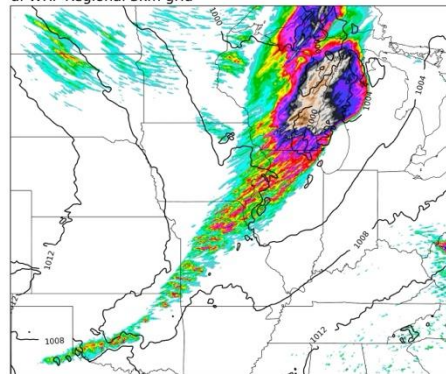


c. MPAS Regional 20km grid

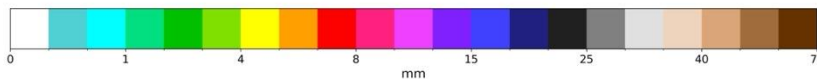
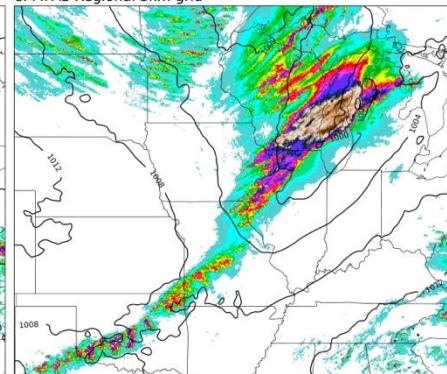


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d. WRF Regional 3km grid



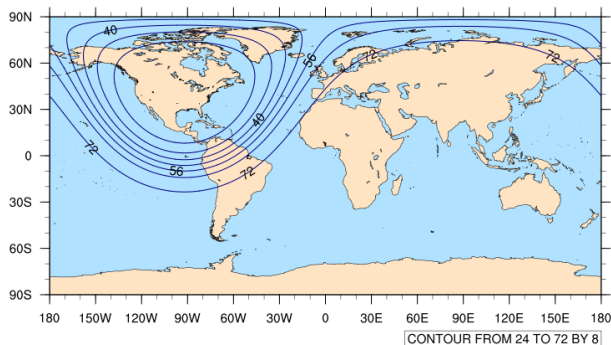
e. MPAS Regional 3km grid



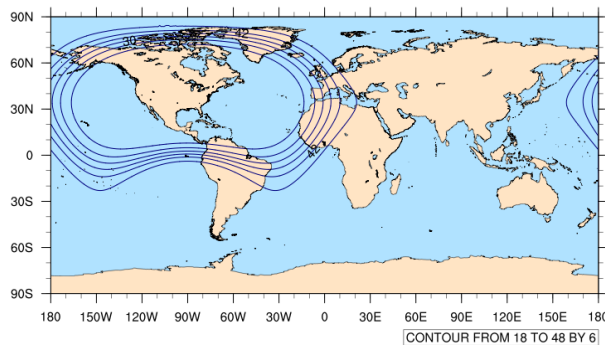
What about different kinds of MPAS domains?

For more fun (!) I ran the case over three different globally refined meshes

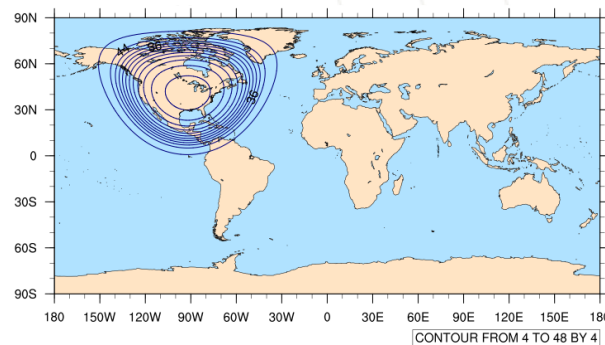
- Meshes: 92-25km, 60-15km and 60-3km. Already existing grids, just used grid_rotate to put them where I wanted.
- It was extremely easy to do the same run over different meshes. Just had to change the physics suite, resolution, graph.info filename and timestep. Four lines!



92-25km

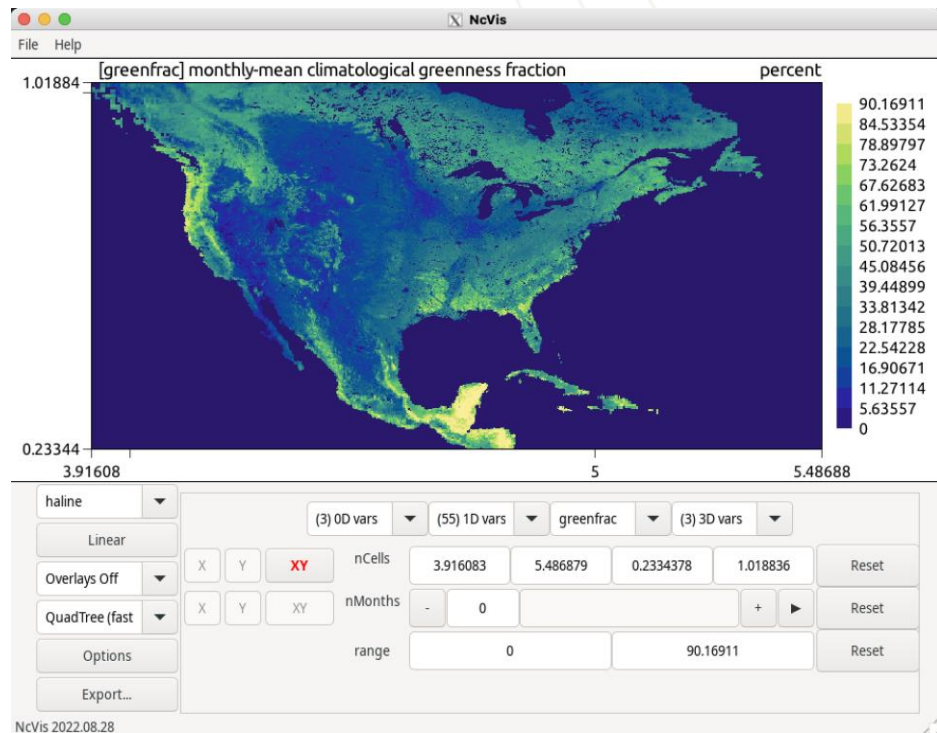
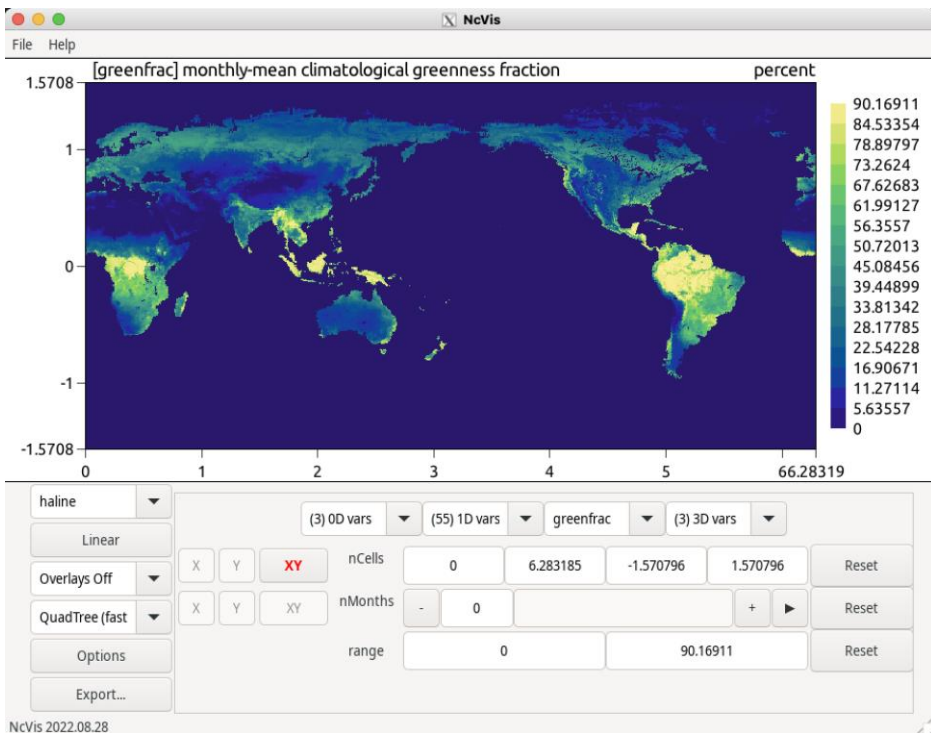


60-15km



60-3km

What about different kinds of MPAS domains?

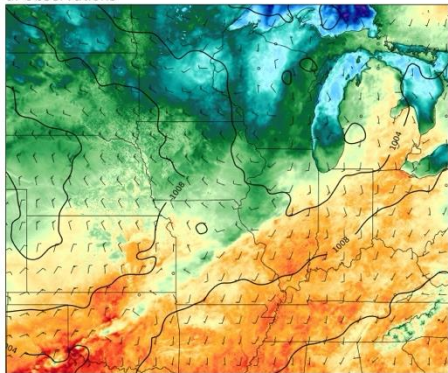


MPAS June 11 7pm Central Time Max 2mTemp

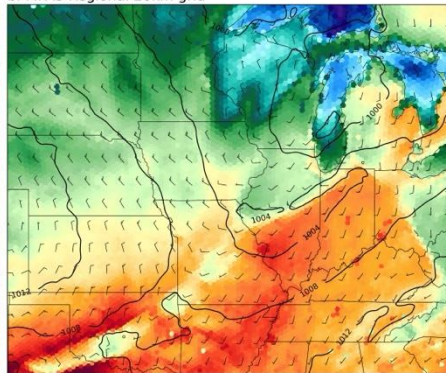


MSLP (hPa) 2m Temp (°C) and Wind (kt)

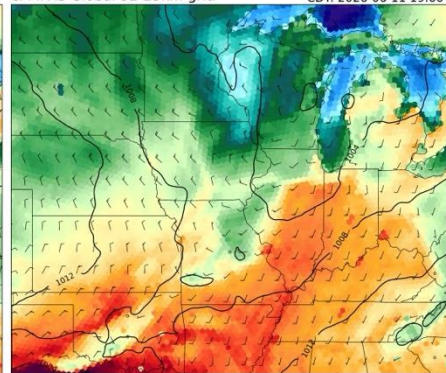
a. Observations



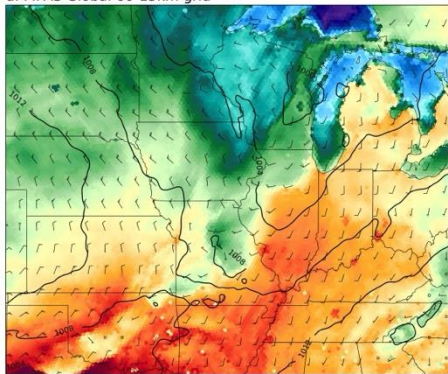
b. MPAS Regional 20km grid



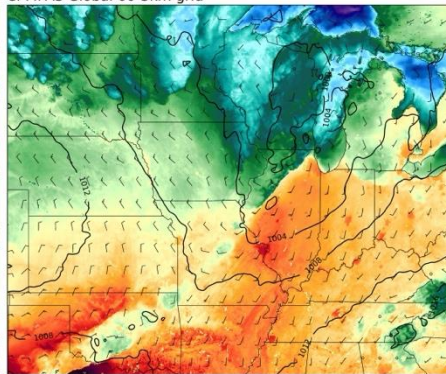
c. MPAS Global 92-25km grid



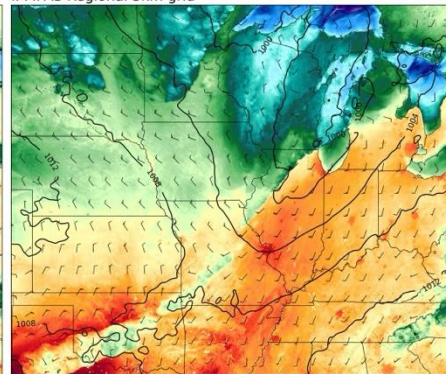
d. MPAS Global 60-15km grid



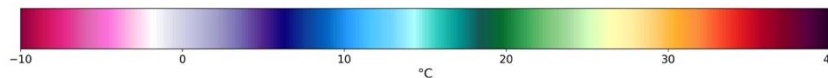
e. MPAS Global 60-3km grid



f. MPAS Regional 3km grid



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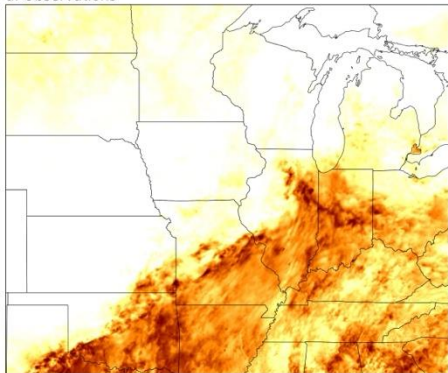


MPAS June 11 7pm Central Time Max CAPE

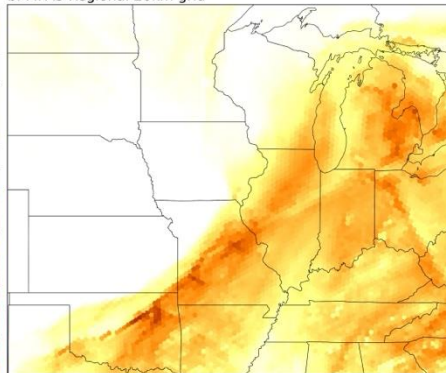


CAPE (J/kg)

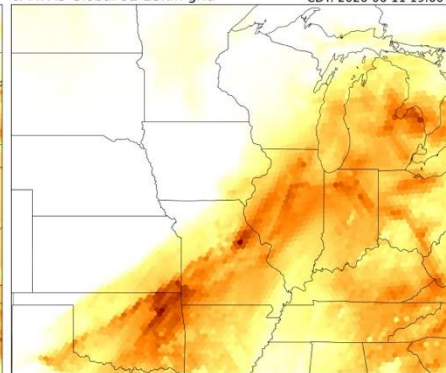
a. Observations



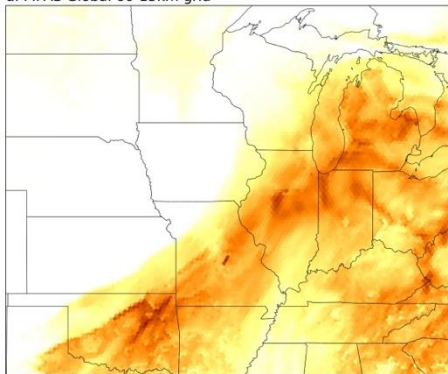
b. MPAS Regional 20km grid



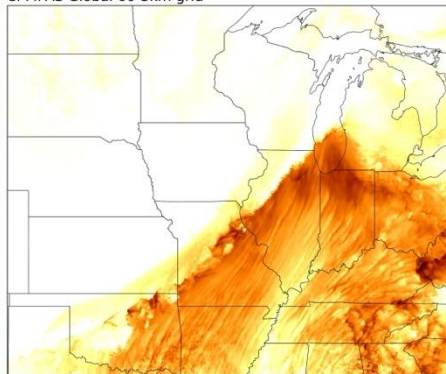
c. MPAS Global 92-25km grid



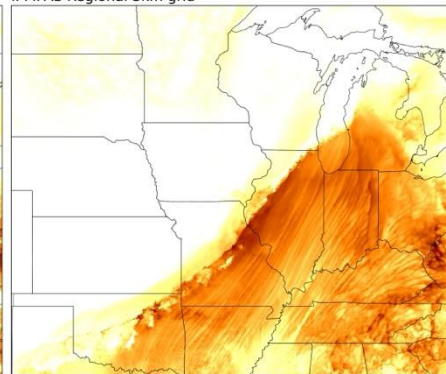
d. MPAS Global 60-15km grid



e. MPAS Global 60-3km grid



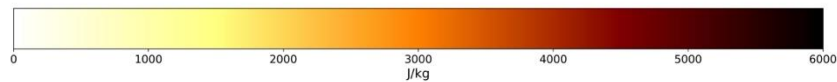
f. MPAS Regional 3km grid



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Valid: 2026-06-12 00:00

CDT: 2026-06-11 19:00

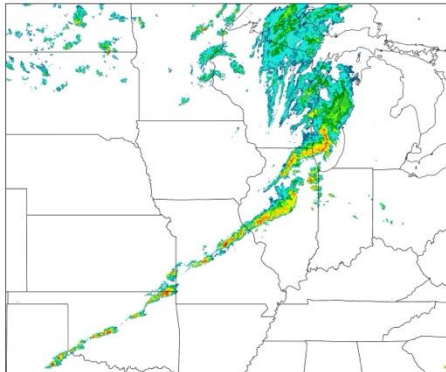


MPAS June 11 7pm Central Time Max dBz



Radar Reflectivity (dBz)

a. Observations



b. MPAS Regional 20km grid

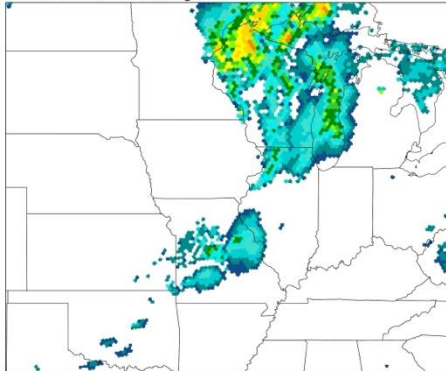


c. MPAS Global 92-25km grid

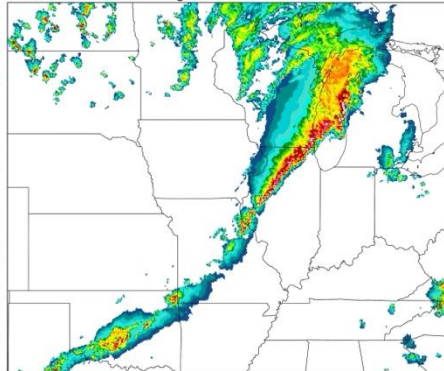


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Valid: 2026-06-12 00:00
CDT: 2026-06-11 19:00

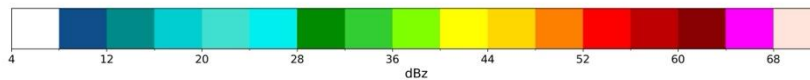
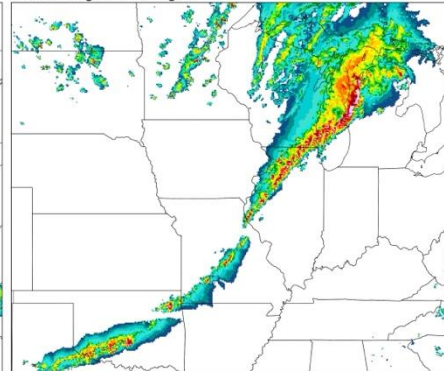
d. MPAS Global 60-15km grid



e. MPAS Global 60-3km grid



f. MPAS Regional 3km grid

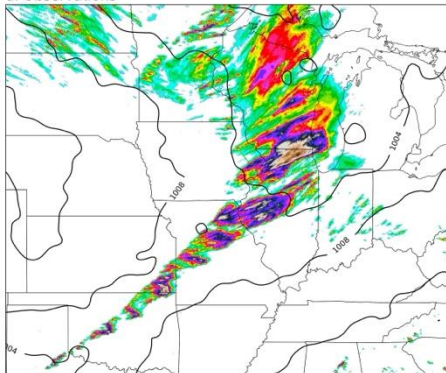


MPAS June 11 7pm Central Time 3hr acc Prec

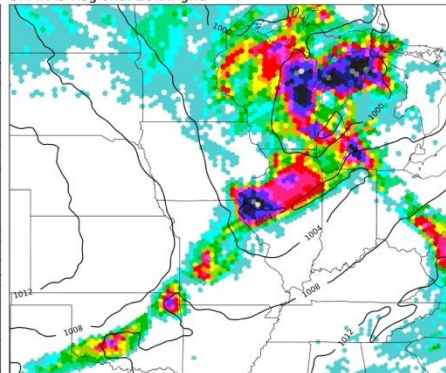


MSLP (hPa) and 3hr Accum. Precip. (mm)

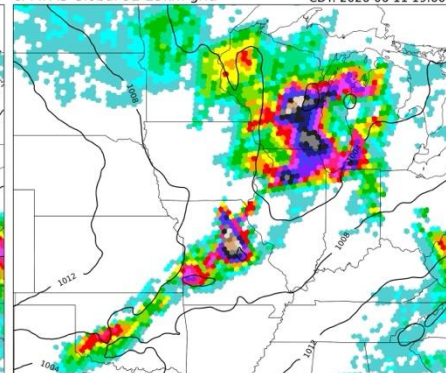
a. Observations



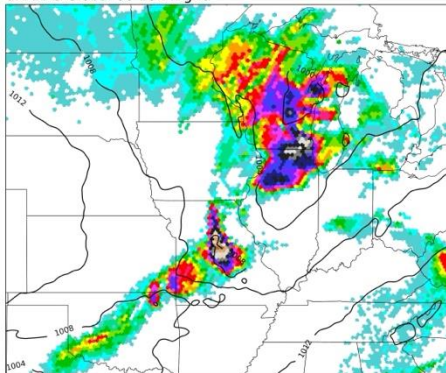
b. MPAS Regional 20km grid



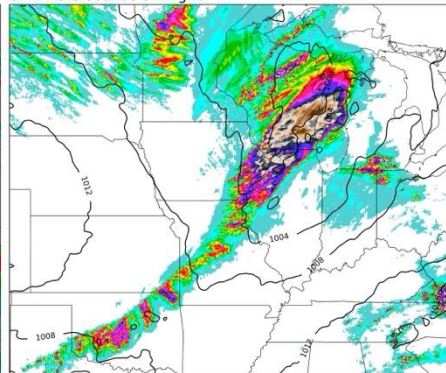
c. MPAS Global 92-25km grid



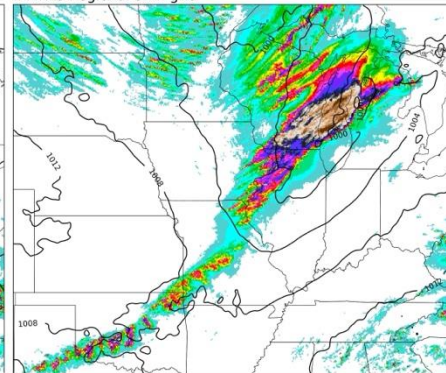
d. MPAS Global 60-15km grid



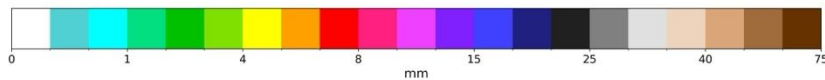
e. MPAS Global 60-3km grid



f. MPAS Regional 3km grid



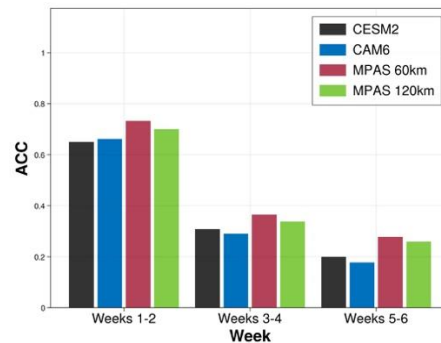
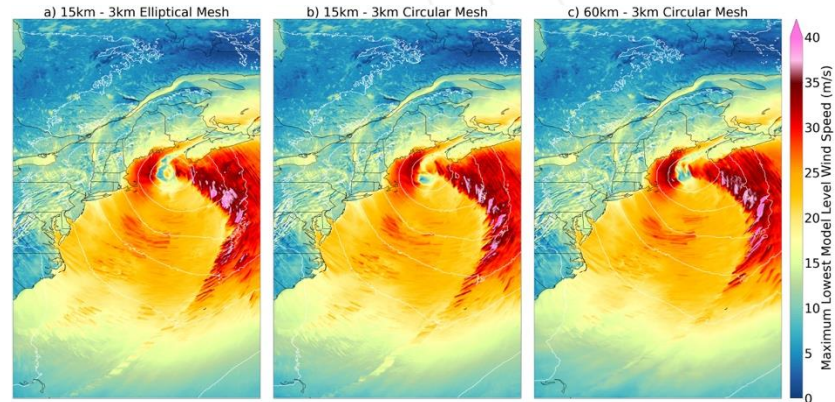
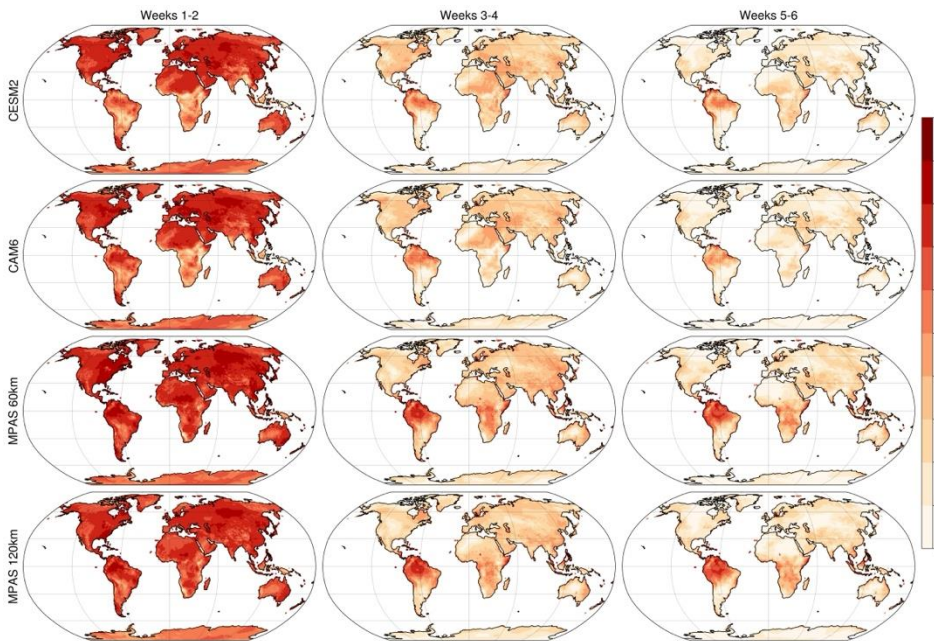
Init: 2026-06-10 12:00
Valid: 2026-06-12 00:00
CDT: 2026-06-11 19:00



Other MPAS stuff I do



- 24 years of weekly initialized S2S hindcasts!
- Multiple meshes ranging from global uniform 120km down to refined resolution of 4km



- Downscaled 600 ETCs from LENS2
- 200 each from early-century, mid-century and late-century

It's not so bad!

- MPAS offers many of the same things WRF has and many more
- Much more flexibility in outputs, diagnostics, etc
- You really can make any kind of mesh you want!
- There are many more talks to come about tools for MPAS mesh creation, analysis, visualization, general post-processing and on and on

Also! Thank you to Kevin Goebbert from Valparaiso University for the WRF namelists!

Namelists and scripts are available

- I have made available, on github, all the python scripts, environments, namelists, etc that I used on the WRF-MPAS comparison I showed today

<https://github.com/abjaye/wrfvmpas>





Questions??

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